Michael He

AI ENGINEER

+1-672-381-3688 | mhe41981@gmail.com | https://www.linkedin.com/in/xinnan-he-827435274/ | https://github.com/MiaoNan | https://miaonan.github.io/mhe_eportfolio/ | Ottawa, ON

Professional Summary

AI engineer with a Master of Applied Data Science, 2 years of machine learning research experience, and 12 months of experience as an AI agent software engineer at Ciena. Built multi-agent workflows, MCP tool integrations, retrieval systems, and internal applications for engineering data and debugging workflows.

Work Experience

Software Engineer and Tools Developer Co-op September 2025 - August 2026

Ciena *Ottawa, Ontario, Canada*

- Developed an orchestration agent for a deployed 10-agent AI debugging system, coordinating sequential and parallel execution, aggregating and evaluating results, handling errors, and creating Jira issues through MCP; supported 4 product teams analyzing over 100 failed test cases.
- Developed Milvus-based RAG agents for Confluence pages, product manuals, and state-dump log analysis, optimizing log retrieval with timestamp-based metadata filtering to support root-cause investigation.
- Optimized RAG pipelines within AI agent applications using reranking, Drain3 log parsing, metadata enrichment, and Neo4j-based knowledge graphs, improving Jira matching and log relevance.
- Prepared LLM training data from 10,000 Confluence pages using reusable tools for cleaning, image-to-text conversion, and question-answer generation, removing approximately 20% of pages as low-value content and describing 3,000 embedded images with Qwen3-6B.

Graduate Student Researcher September 2021 - September 2023

Chongqing University *Chongqing, China*

- Built, trained, and evaluated a PyTorch Transformer to predict keywords in future book titles using a dataset of 20,000 historical titles.
- Modified, trained, and evaluated a PyTorch Faster R-CNN model with domain adaptation techniques to reduce feature distribution differences and improve cross-domain object detection mAP.

Data Analyst Intern December 2020 - February 2021

Tianbojincheng Intelligent Medical Technology (Nanjing) *Nanjing, China*

- Built and trained a PyTorch deep learning model to fill missing values in a dataset containing 20,000 gene expression features across 40 tissue regions, supporting downstream data analysis.

Education

Master of Applied Data Science September 2024 - August 2026

University of Victoria

Bachelor of Computer Science and Technology September 2017 - June 2021

Hohai University

Technical Skills

Programming: Python, JavaScript, SQL
Applied AI: Multi-agent systems, Google ADK, MCP tool integration, RAG, Milvus, Neo4j, LLM training-data preparation
Software Engineering: FastAPI, Flask, REST APIs, React, Git, Linux, Gunicorn, Docker
Machine Learning & Data: PyTorch, scikit-learn, spaCy, model evaluation, data preprocessing

Awards & Honors

Runner-up among more than 200 teams at the 2026 Ciena AI Hackathon. Our 5-person team, Causal Chain, built an AI-powered debugging pipeline.

First-Class Scholarship (Category A), Chongqing University; Academic Excellence Scholarship, Hohai University